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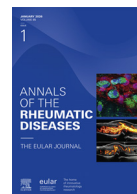
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Myositis

Multiomics analysis uncovers subtype-specific mechanisms and biomarkers in idiopathic inflammatory myopathies

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ARTICLE INFO

Article history:

Received 25 March 2025

Received in revised form 22 July 2025

Accepted 8 August 2025

ABSTRACT

Objectives: Idiopathic inflammatory myopathies (IIM) are autoimmune disorders with distinct subtype features, but their molecular mechanisms remain unclear. This study integrated multiomics data to identify subtype-specific molecular signatures and evaluate their prognostic significance in a Han Chinese IIM cohort.

Methods: RNA sequencing, proteomics, and metabolomics were generated on muscle tissues from 203 patients with IIM (including 44 in a validation cohort) and 18 controls. Differential expression was analysed for exons, intron retentions (IRs), proteins, and metabolites, integrated via multiomics factor analysis (MOFA). Pathway enrichment, single-sample Gene Set Enrichment Analysis (ssGSEA), correlation with clinical features, receiver operating characteristic curve, and survival analyses were conducted.

Results: MOFA distinguished dermatomyositis (DM), immune-mediated necrotising myopathy (IMNM), and antisynthetase syndrome (ASyS) from controls, identifying 798, 748, and 297 subtype-specific features and pathways, respectively, which were further validated in an independent cohort. In DM, upregulated interferon (IFN) and cytokine pathways were prominent, with 11 IFN-related genes altered at exon, IR, and protein levels, alongside changes in related metabolites. IFNs and cytokine scores correlated with skin manifestations, perifascicular atrophy/necrosis, inflammation, and relapse risk. IMNM showed changes in myosin, actin, and troponin genes, with enrichment of cytoskeleton and extracellular matrix (ECM) pathways that were positively linked to muscle necrosis, regeneration, and inflammation. Protein-level of ECM-related pathways predicted a favourable prognosis. ASyS displayed distinct metabolic signatures

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Handling editor Josef S. Smolen.

<https://doi.org/10.1016/j.ard.2025.08.011>

(nucleosides, ketones, phosphatidylserine) and endothelial dysfunction, with key metabolism-regulated genes (*FABP3*, *AKR1C2*, *AKR1C3*) showing multiomics alterations associated with necrosis, inflammation, and prognosis.

Conclusions: This multiomics analysis elucidates distinct molecular mechanisms in IIM subtypes, identifying potential biomarkers for personalised prognosis and therapy.

WHAT IS ALREADY KNOWN ON THIS TOPIC?

- Idiopathic inflammatory myopathies (IIM)-affected muscles exhibit distinct pathogenic characteristics across dermatomyositis (DM), antisynthetase syndrome (ASyS), immune-mediated necrotising myopathy (IMNM), and other subtypes.
- Transcriptomic analysis of muscle tissues has identified distinct gene expression changes across IIM subtypes.
- An upregulated interferon gene signature has been observed in DM, while plasma metabolite alterations have been reported in ASyS.

WHAT THIS STUDY ADDS

- Integrative multiomics analysis using multiomics factor analysis (MOFA) provides a powerful framework, overcoming the limitations of single-omics approaches.
- MOFA identifies distinct molecular pathways in IIM subtypes, linking IFN and cytokine signalling to DM, cytoskeletal/extracellular matrix disruption to IMNM, and metabolic/vascular alterations to ASyS.
- IIM subtype-specific pathways strongly correlate with clinical characteristics and play a predictive role in IIM disease progression.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY?

- This multiomics analysis reveals distinct molecular mechanisms in IIM subtypes, identifying potential biomarkers for personalised prognosis and therapy.

INTRODUCTION

Idiopathic inflammatory myopathies (IIM) are a diverse group of systemic autoimmune disorders. The primary clinical feature is muscle weakness, but other organs, including the skin, joints, lungs, heart, and gastrointestinal tract, can also be affected [1]. The subtypes of IIM include dermatomyositis (DM), juvenile dermatomyositis (JDM), antisynthetase syndrome (ASyS), immune-mediated necrotising myopathy (IMNM), polymyositis, inclusion body myositis (IBM), and overlap myositis [2].

Organ involvement, treatment responses, and prognosis vary among these subtypes, suggesting distinct pathophysiological mechanisms for each. Notably, DM and ASyS often involve multiple organs, including the skin, lungs, and/or joints, while IMNM and IBM primarily affect skeletal muscle [1]. Muscle biopsy is crucial for diagnosing IIM, as it helps identify subtype-specific features such as perifascicular atrophy and perivascular inflammatory infiltrates in DM; perifascicular myofiber necrosis, perifascicular atrophy, and perimysial fragmentation in ASyS; necrotic muscle fibres and major histocompatibility complex (MHC-I) hyperexpression on the sarcolemma of muscle fibres in IMNM; and lymphocytic and macrophage aggregates in the endomysium, along with rimmed vacuoles in IBM [3].

Several transcriptomic analyses of muscle tissue have revealed distinct gene expression changes across IIM subtypes. The interferon (IFN) pathway, for instance, displays variable activation levels among subtypes—being predominantly activated in DM and minimally activated in IMNM [4,5]. A co-expression network analysis of dynamically regulated transcripts further organised the myositis transcriptome into gene expression modules, identifying subtype-specific signatures such as vasculogenesis in IBM and RNA processing in ASyS, which contribute to molecular subclustering of patients with IIM [6]. Additionally, anti-Mi2 DM exhibited a distinct gene expression profile compared with other IIM subtypes. These genes were associated with the clinical features of patients with anti-Mi2 DM and suggest a possible pathogenic role of the anti-Mi2 antibody [7].

However, the distinctive pathways involved in IIM subtypes remain largely unexplored, and their associations with clinical indices are still unclear. Moreover, ethnicity plays a significant role in the prevalence, clinical phenotype, and muscle pathology of IIM. For example, IBM is exceedingly rare in Han Chinese, representing only 0.68% of all neuromuscular disorders, whereas it is the most common myopathy in individuals over 40 years old in Caucasian populations [8,9]. Similarly, the prevalence and severity of muscle involvement in anti-MDA5 DM vary across populations [1,10,11]. Moreover, the use of any immunosuppressant may also affect the muscle tissue transcriptome in previous studies [12]. In this study, we analysed multiomics data, including transcriptomics (exon and intron retention [IR]), proteomic, and metabolomic profiles, from muscle samples of 159 patients with treatment-naïve IIM in Han Chinese using an unsupervised method called ‘multiomics factor analysis (MOFA)’ [13,14]. MOFA integrated multiomic data to identify key features (exons, IRs, proteins, or metabolites) specific to each IIM subtype. Additionally, we performed pathway enrichment analysis of these identified features and examined their correlations with clinical indices and disease prognosis. Our study provides valuable insights into the pathogenic mechanisms underlying muscle injury in different IIM subtypes, offering critical information for understanding disease pathology and identifying potential therapeutic targets.

MATERIALS AND METHODS

Patient recruitment

Patients with IIM were recruited from Xiangya Hospital, Central South University, between 2016 and 2023. The 2018 European Neuromuscular Centre (ENMC) criteria, 2016 ENMC criteria and Connor criteria were used to determine DM, IMNM, and ASyS, respectively [15–17]. All patients were newly diagnosed with IIM and had not received glucocorticoid or immunosuppressant treatment. IBM was not included due to its extreme rarity in China [8]. Patients with infections, malignancies, overlap syndromes, severe cardiovascular or metabolic conditions, or those who had received immunosuppressants prior to biopsy were excluded.

Myositis-specific antibodies (MSAs) were tested using Euro-line Myositis Profile immunoblot (detecting 18 MSAs, Euroimmun, Germany). Based on MSAs, patients were further grouped into anti-MDA5, anti-Mi2, anti-SAE, anti-NXP2, anti-TIF1 γ , anti-Jo-1, anti-EJ, anti-PL-7, anti-PL-12, anti-SRP, anti-HMGCR, and MSAs-negative subgroups. All patients with IIM were assessed for the severity of myasthenia using the Manual Muscle Test-8 (MMT-8) on neck flexors, deltoid, biceps, gluteus maximus, gluteus medius, quadriceps, wrist extensors, and ankle dorsiflexors. Specially, electroneuromyography further validated myogenic damage in the right biceps brachii muscle. A skeletal muscle biopsy was performed on the medial aspect of the left biceps brachii for diagnostic purposes. Normal control muscle biopsies were obtained from patients undergoing knee or shoulder joint replacement surgeries, and were excluded with infection, tumour, severe cardiovascular, metabolic disease or those who had received immunosuppressants.

All muscle biopsy samples were subjected to RNA sequencing, mass spectrometry, and untargeted metabolomics analysis. After quality control, a total of 203 patients with IIM and 18 normal controls had qualified transcriptomic, proteomic, or metabolomic data. For MOFA analysis, we included patients with IIM with a single strongly positive MSA and complete transcriptomic, proteomic, and metabolomic data post-quality control. This resulted in a cohort of 159 patients with IIM and 18 normal controls ([Supplementary Table S1](#)). Additionally, a validation cohort of 44 patients with IIM, including 29 MSA-negative cases, was incorporated to confirm the findings across the 4 multi-omics datasets. Detailed information on the validation cohort is summarised in [Supplementary Table S2](#).

This research was reviewed and approved by the ethics committee of Xiangya Hospital, Central South University (201212074). Before their involvement in the study, all participants signed an informed consent form. The design, conduct, reporting, and dissemination plans of this research were not influenced by patients or the public.

Statistical analysis

All data analyses were performed using R software (V.4.3.0). Data normality was assessed using the Shapiro-Wilk test. Quantitative variables were reported as median with interquartile range. For 2 independent samples with normally distributed data and homogeneous variances, an unpaired t-test was used; the Mann-Whitney test was applied when variances were unequal. For multiple comparisons of patients' general and clinical data, analysis of variance was conducted on normally distributed data with homogeneous variances, followed by Tukey's post hoc test. For skewed data, or normally distributed data lacking homogeneity of variance, the Kruskal-Wallis test was performed, followed by Dunn's post hoc test. *P* values from post hoc analyses were adjusted using the Benjamini-Hochberg method. Chi-square and Fisher's exact tests were applied to categorical variables. Correlation analyses were conducted using Pearson's correlation for normally distributed variables, Spearman's correlation for non-normally distributed variables, and the point-biserial correlation test for relationships between continuous and categorical variables. Cox proportional hazards model was used to assess the correlation between pathway activity, metabolite expression and patient prognosis. A 2-sided *P* value <.05 was considered statistically significant.

Detailed methods for RNA sequencing, mass spectrometry (MS) analysis, untargeted metabolomics, pathway analysis,

single-sample Gene Set Enrichment Analysis (ssGSEA), receiver operating characteristic (ROC) curve analysis, and survival analysis are provided in the [Supplementary Materials](#).

RESULTS

Key molecular determinants for IIM subtype classification identified through integrative multiomics analysis

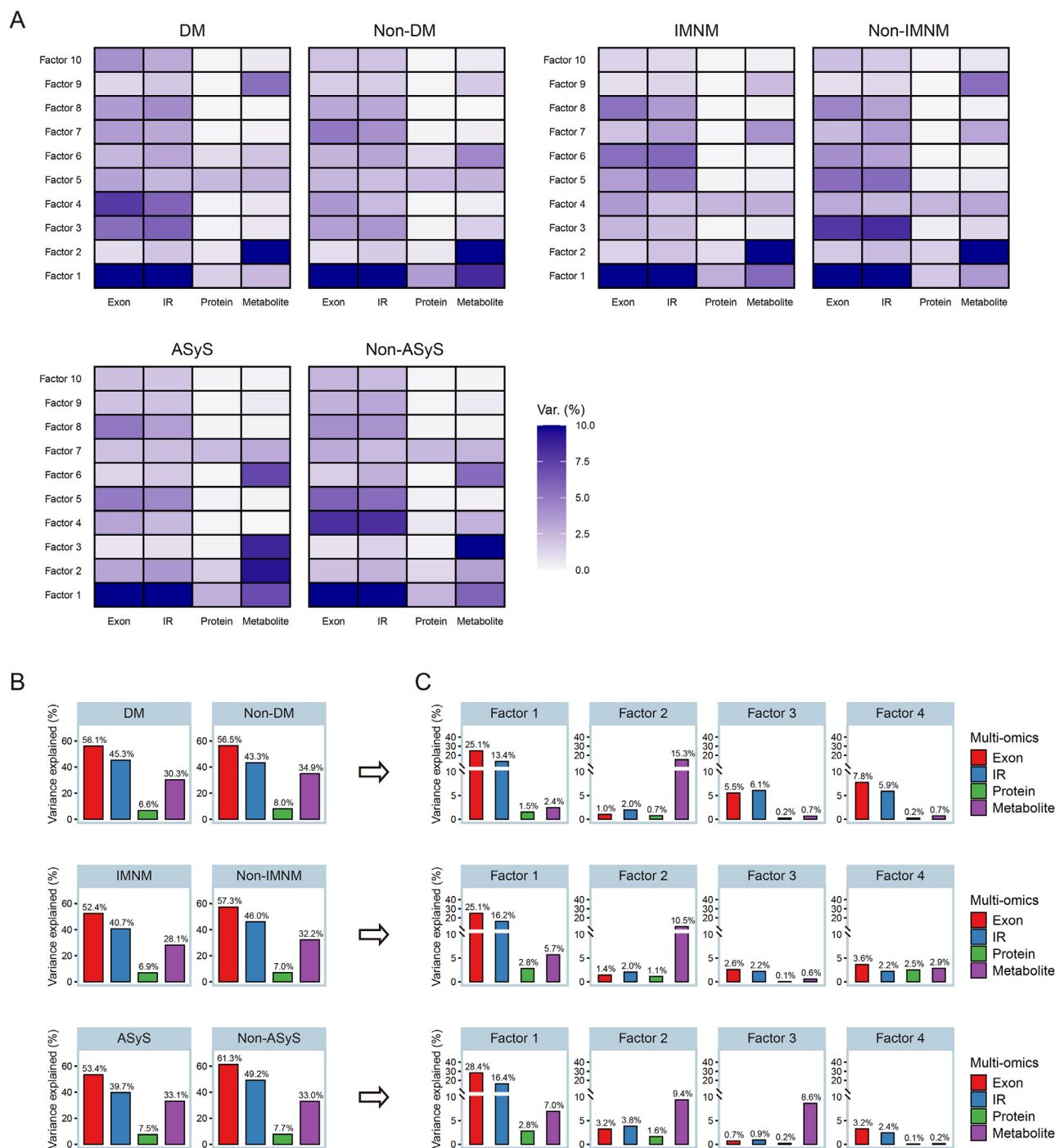
In total, after quality control, we obtained muscle sample data from 159 patients with newly diagnosed IIM and 18 controls, covering 4 types of omics data: genomic exons, IRs, proteomic, and metabolomic profiles. Using a significance threshold of *P* < .05, compared with the control group, we identified 7872 differentially expressed exons (DEEs; 7250 in DM, 5379 in IMNM, and 4034 in ASyS), 1490 differentially expressed IRs (DEIs; 1166 in DM, 1076 in IMNM, and 836 in ASyS), 3697 differentially expressed proteins (DEPs; 2509 in DM, 2209 in IMNM, and 2367 in ASyS), and 1137 differentially expressed metabolites (DEMs; 976 in DM, 851 in IMNM, and 973 in ASyS). Principal component analysis (PCA) of the top 1500 DEEs and DEPs, as well as all DEIs and DEMs, revealed that while the IIM group could be distinguished from the control group, the subtypes within the IIM group (DM, IMNM, and ASyS) were not clearly separated ([Supplementary Fig S1](#)). This suggests that single-omics data alone are insufficient for differentiating between IIM subtypes.

To address this limitation, we conducted an integrative analysis of the 4 omics datasets using MOFA [13]. This approach identified 10 factors for each comparison (DM vs non-DM, IMNM vs non-IMNM, and ASyS vs non-ASyS), with each explaining at least 1% of the variance in at least 1 omics dataset and representing a distinct combination of exons, IRs, proteins, and metabolites ([Fig 1A](#)). Among them, factor 1 contributed the most to distinguishing IIM subtypes and was primarily driven by exons and IRs, with smaller influences from proteins and metabolites. Notably, in the DM vs non-DM comparison, the model explains 56.1% of the variance in exon data, 45.3% in IR data, 6.6% in proteomic data, and 30.3% in metabolomic data across 10 factors ([Fig 1B](#)). To explore this further, we examined how individual omics features contributed to the variance in these 4 factors. The influence varied across groups. For example, in DM, exons accounted for 25.1% of factor 1, followed by IRs (13.4%), proteins (1.5%), and metabolites (2.4%). In factor 2, the proportions shifted to 1.0%, 2.0%, 0.7%, and 15.3%, respectively, highlighting distinct contributions from different data modalities ([Fig 1C](#)).

To better evaluate the model's ability to distinguish IIM subtypes, we performed ROC curve analysis using feature genes derived from MOFA factors 1–4. The ROC curves demonstrated comparable classification performance, with the area under the curve (AUC) values of 78.4% for DM, 82.2% for IMNM, and 79.1% for ASyS. The corresponding specificities were 81.3% for DM, 88.9% for IMNM, and 90.2% for ASyS ([Supplementary Fig S2](#)).

Distinct features and pathways in IIM subtypes

To identify the most critical molecular features in each IIM subtype, we extracted the top 1800 features from factor 1 and the top 400 features from factors 2, 3, and 4, respectively. Integrating these multimodal features revealed clear separation among IIM subtypes (DM, IMNM, ASyS) and controls in the PCA plot, indicating that the combined data effectively capture the defining characteristics of each subtype ([Fig 2A](#)). Specifically, we identified

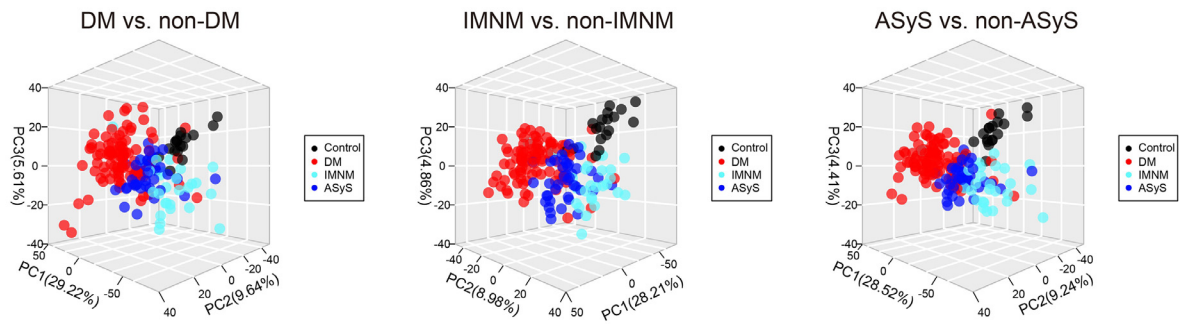


344 upregulated and 454 downregulated features associated with DM. In IMNM, 748 subtype-specific features were detected, including 597 upregulated and 151 downregulated ones. For ASyS, we identified 297 features, with 146 upregulated and 151 downregulated (Fig 2B). Pathway enrichment analysis highlighted distinct molecular signatures for each IIM subtype (Fig 2C). In DM, the most prominent pathways involved IFN-related signalling, cytokine-mediated signalling, and the I kappa B kinase (IKK), nuclear factor-kappa B (NF- κ B) IKK/NF- κ B pathway. IMNM was characterised by enrichment in cytoskeleton-

and extracellular matrix (ECM)-related processes, including actin filament bundle assembly, ECM organisation, and tissue remodeling, along with pathways linked to macrophage migration, myeloid cell and T cell differentiation and natural killer-T (NK-T) cell differentiation. In ASyS, metabolism- and vascular-related pathways (e.g. endothelial cell differentiation and endothelium development) were predominant, with notable enrichment in antigen processing and presentation pathways (Fig 2C).

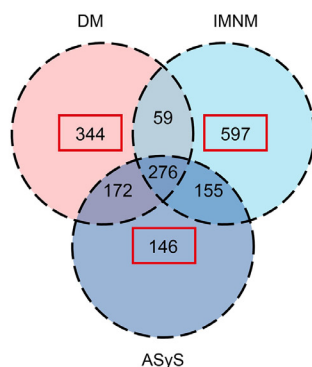
Considering the diversity of IIM MSA subtypes and the heterogeneity of inflammatory infiltration in muscle biopsies, we

A Multi-omics integrated analysis

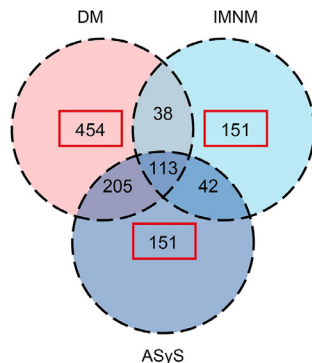


B

Up-regulated exons, IRs, proteins, and metabolites



Down-regulated exons, IRs, proteins, and metabolites



C

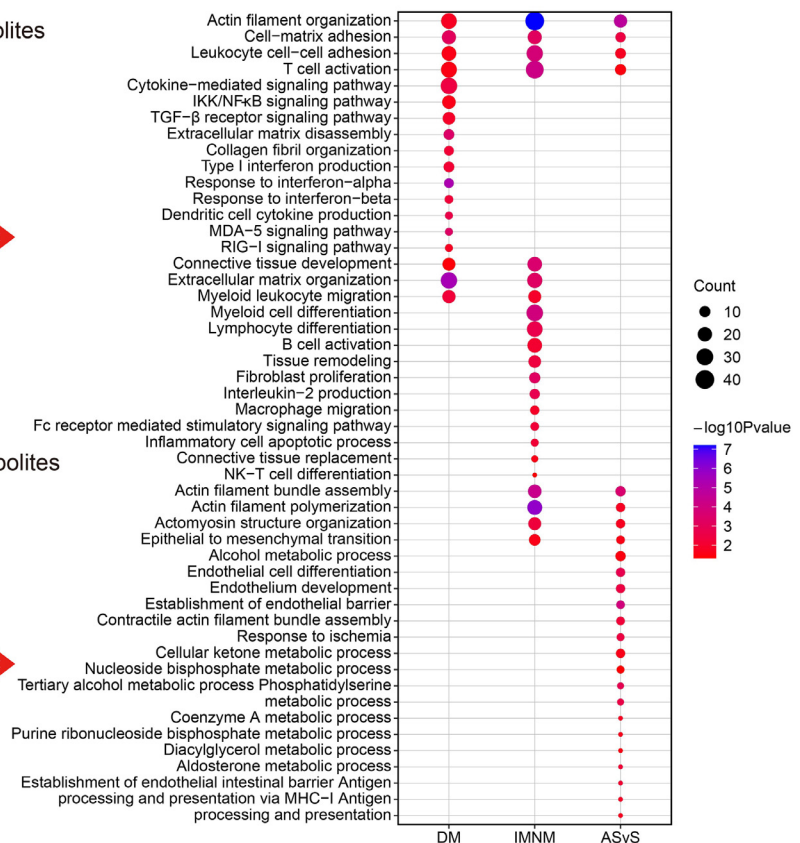


Figure 2. Distinct features and pathways in IIM subtypes. (A) 3D PCA plot of IIM subtypes based on the top 1800 features (exons, IRs, proteins, metabolites) from factor 1 and the top 400 features from factors 2–4 in MOFA. Dot colours represent IIM subtypes. (B) Venn diagram showing the overlap of features among IIM subtypes, with nonoverlapping features considered subtype specific. (C) Bubble plot illustrating biological pathways enriched by IIM subtype-specific features, where dot size represents the number of features and dot colour indicates *P* values. ASyS, antisynthetase syndrome; DM, dermatomyositis; IIM, idiopathic inflammatory myopathy; IMNM, immune-mediated necrotising myopathy; IRs, intron retentions; MOFA, multiomics factor analysis; PCA, principal component analysis.

also performed MOFA on the following groups: MSA subgroups (anti-MDA5, anti-Jo-1, anti-TIF1 γ , and anti-SRP), IIM subtypes (DM, IMNM, and ASyS) in biopsies with inflammatory infiltration, IIM subtypes (DM and ASyS) in noninflamed muscle biopsies, and anti-MDA5 DM with and without inflammatory infiltration. Detailed results are provided in the [Supplementary Materials and Supplementary Figs S3–8](#).

Characteristic alterations in IFN and cytokine pathway in DM

We conducted a detailed analysis to assess whether multiomics data collectively contribute to IIM subtype-specific pathways. In DM, we identified 73 features involved in IFN-related pathways, including 33 linked to IFN- α/β production and 40

associated with IFN- γ production ([Fig 3A](#), [Supplementary Figs S9 and S10](#)). Nearly all genes exhibited significant exon-level changes, with alterations also observed in 19 IRs and 25 proteins. Notably, 11 genes (DHX58, MX2, ISG15, MX1, STAT1, VIM, HLA-A, PARP9, LGALS9, PARP14, and TRIM21) displayed changes across exons, IRs, and proteins. DHX58 showed significant upregulation at the exon and IR levels but a decrease in protein expression, suggesting a potential negative regulatory role for its IRs. Given that DHX58 inhibits RIG-I signalling, its reduced protein levels in DM may contribute to increased type I IFN production ([Fig 3A](#)).

Cytokine-mediated pathway was also significantly enriched in DM, involving 74 genes ([Fig 3B](#), [Supplementary Fig S11](#)). Most of these genes showed significant exon-level alterations,

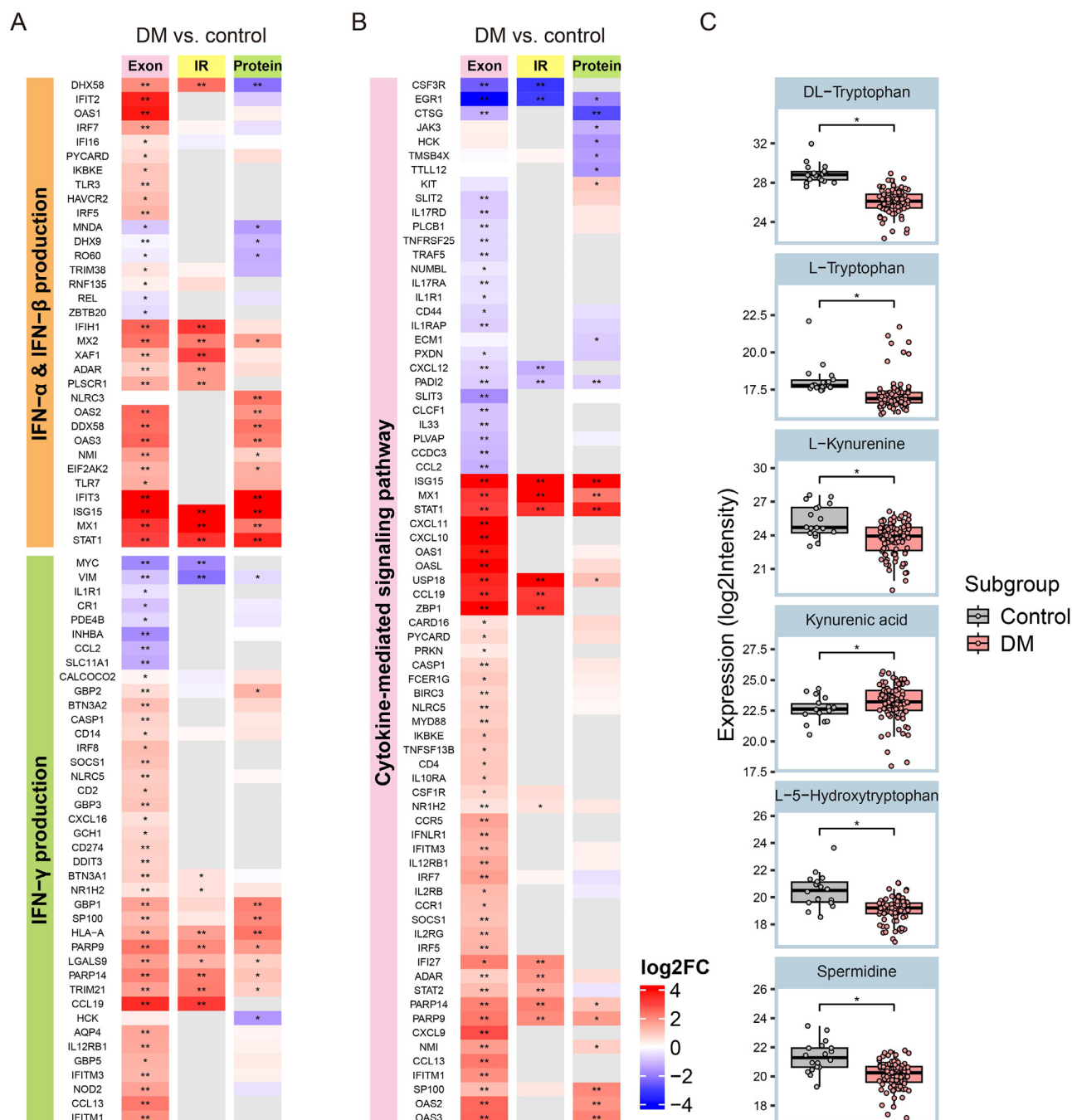


Figure 3. Interferon and cytokine pathway alterations in DM. (A) Heatmap depicting exon, IR, and protein expression of genes associated with IFN- α/β ($n = 33$) and IFN- γ ($n = 40$) production. (B) Heatmap showing the expression of 74 genes involved in cytokine-mediated signalling. The colour gradient represents the $\log_2(\text{fold change})$, with red indicating higher expression. (C) Box plots showing alterations of 6 IFN-related metabolites in DM. Statistical significance: * $P < .05$, ** $P < .001$ (DM vs control). DM, dermatomyositis; IFN, interferon; IRs, intron retentions.

with 16 IRs and 19 proteins also exhibiting notable changes. Eight genes displayed simultaneous alterations at the exon, IR, and protein levels. Notably, cytokine genes such as CCL19, CXCL9-11, and CCL13 primarily exhibited exon-level changes, yet their protein levels were undetectable in the vast majority of samples. This may be attributed to several factors, including their very low abundance, the timing of muscle biopsy, and the inherent instability and short half-life of these small proteins.

We identified alterations in IFN-related metabolites, including tryptophan (DL-tryptophan and L-tryptophan), whose metabolites suppress stimulator of interferon gene (STING)-dependent type I IFN production [18] and spermidine, which inhibits IFN- α expression in plasmacytoid dendritic cells [19];

both were decreased in DM. In contrast, kynurenine metabolites (kynurenic acid), which are IFN- γ inducible biomarkers, were increased in DM (Fig 3C) [20].

To better illustrate these molecular alterations in DM and their upstream-downstream relationships, we mapped them in pathway diagrams for IFN and cytokine signalling (Supplementary Figs S12 and S13).

To further validate the above results, we analysed exon, IR, protein, and metabolite expression in the validation cohort of 30 patients with DM. IFN and cytokine pathways showed 80.3% consistency with previous findings at $P < .05$ vs control (91.8% demonstrating consistent directional changes). Metabolite data reflected these same patterns (Supplementary Fig S14).

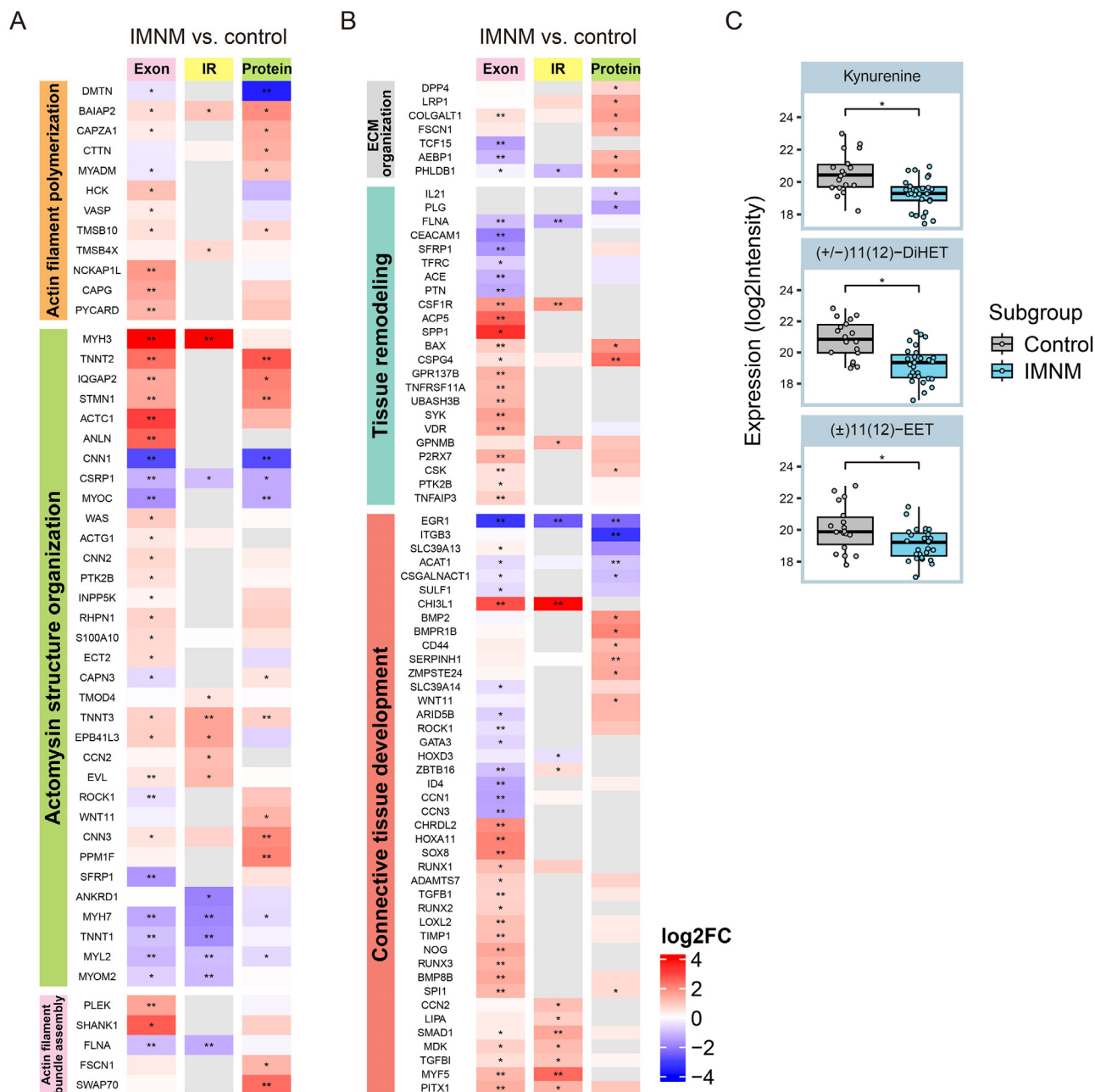


Figure 4. Actin filament- and extracellular matrix (ECM)-related pathway changes in IMNM. (A) Heatmap showing the exon, IR, and protein expression of 12 genes involved in actin filament polymerisation, 33 genes in actomyosin structure organisation, and 5 genes in actin filament bundle assembly. (B) Heatmap depicting the exon, IR, and protein expression of 7 genes related to ECM organisation, 23 genes involved in tissue remodelling, and 42 genes in connective tissue development. The colour gradient represents log₂(fold change), with red indicating higher expression. (C) Box plots showing alterations of 3 actin filament-related metabolites in IMNM. Statistical significance: * $P < .05$, ** $P < .001$ (IMNM vs control). IMNM, immune-mediated necrotising myopathy; IRs, intron retentions.

Cytoskeleton- and ECM-related pathway changes in IMNM

The most significantly enriched pathways in IMNM were cytoskeleton- and ECM-related pathways. A total of 50 genes (41 exons, 15 IRs, and 21 proteins) were altered in cytoskeleton-related pathways, including 12 involved in actin filament polymerisation, 33 in actomyosin structure organisation, and 5 in actin filament bundle assembly (Fig 4A, Supplementary Fig S15). Notably, BAIAP2, CSRP1, TNNT3, MYH7, and MYL2 exhibited changes at the exon, IR, and protein levels.

Among all exonic changes, 68.3% of genes were upregulated, while 31.7% were downregulated. Within the myosin gene family, MYH7, MYL2, and MYOM2 were downregulated in the exon, whereas MYH3 was upregulated in the exon. Actin-related genes

TMOD4 (IR), FSCN1 (protein), ACTC1 (exon), and ACTG1 (exon) showed increased expression levels. Among troponin genes, TNNT2 (exon, protein) and TNNT3 (exon, IR, protein) were upregulated, while TNNT1 (exon, IR) was downregulated. CAPN3 presented decreased exon and increased protein. These alterations suggest disruptions in cytoskeletal integrity and muscle contractility, potentially indicating compensatory mechanisms or shifts in actomyosin regulation (Fig 4A).

We identified 72 genes (56 exons, 15 IRs, and 22 proteins) enriched in ECM-related pathways, including 7 involved in ECM organisation, 23 in tissue remodelling, and 42 in connective tissue development (Fig 4B, Supplementary Fig S16). PHLDB1 exhibited downregulated exons and IRs, but upregulated protein expression, suggesting negative regulation by IRs. Key signalling

molecules *TGFB1* (exon) and *WNT11* (protein) showed alterations, emphasising their importance in ECM modulation. Seventeen transcription factors were altered in ECM-related pathways in IMNM. *RUNX1*, *RUNX2*, *RUNX3*, *VDR*, *HOXA11*, and *SOX8* showed increased exon expression, while *TCF15*, *ARID5B*, *GATA3*, and *ID4* were decreased. *SMAD1*, *PITX1*, and *MYF5* were upregulated at both exon and IR levels. *SP1* increased at both the exon and protein levels. *ZBTB16* showed opposite changes in exon (down) and IR (up), and *EGR1* was downregulated across all levels. These findings reflect the complex, multi-layered regulation of transcription factors within ECM-associated pathways in IMNM. Additionally, *ADAMTS7*, an ECM-cleaving enzyme, revealed significant exon changes, underscoring its role in ECM remodelling (Fig 4B).

Moreover, cytoskeleton- and ECM-related metabolites, including kynurenine, (±)11(12)-DiHET, and (±)11(12)-EET, were decreased in IMNM compared to controls (Fig 4C).

The expression changes, classification, and relationships of these multimodal features were visualised in the diagrams of cytoskeleton- and ECM-related pathways (Supplementary Figs S17 and S18).

We further analysed exon, IR, protein, and metabolite expression in the validation cohort comprising 10 patients with IMNM. Cytoskeleton- and ECM-related pathways demonstrated 74.6% consistency at $P < .05$ vs control, with 84.4% of changes exhibiting consistent directionality. Metabolite profiles largely mirrored these patterns (Supplementary Fig S19).

Metabolic pathway alterations and endothelial dysfunction in ASyS

Our study revealed significant enrichment of metabolism-related pathways in ASyS, identifying 65 genes across various subcategories (Fig 5A, B, Supplementary Fig S20). These metabolic changes included 11 genes in nucleotide metabolism, 32 genes in ketone metabolism, 4 genes in the phosphatidylserine pathway, and 18 genes in alcohol metabolism. Notably, *FABP3*, *AKR1C2*, *AKR1C3*, and *EGR1* displayed alterations at the exon, IR, and protein levels. *FABP3*, involved in fatty acid transport, along with *AKR1C2* and *AKR1C3*, which are key regulators of steroid and prostaglandin metabolism, together contribute to the modulation of hormonal balance and inflammation.

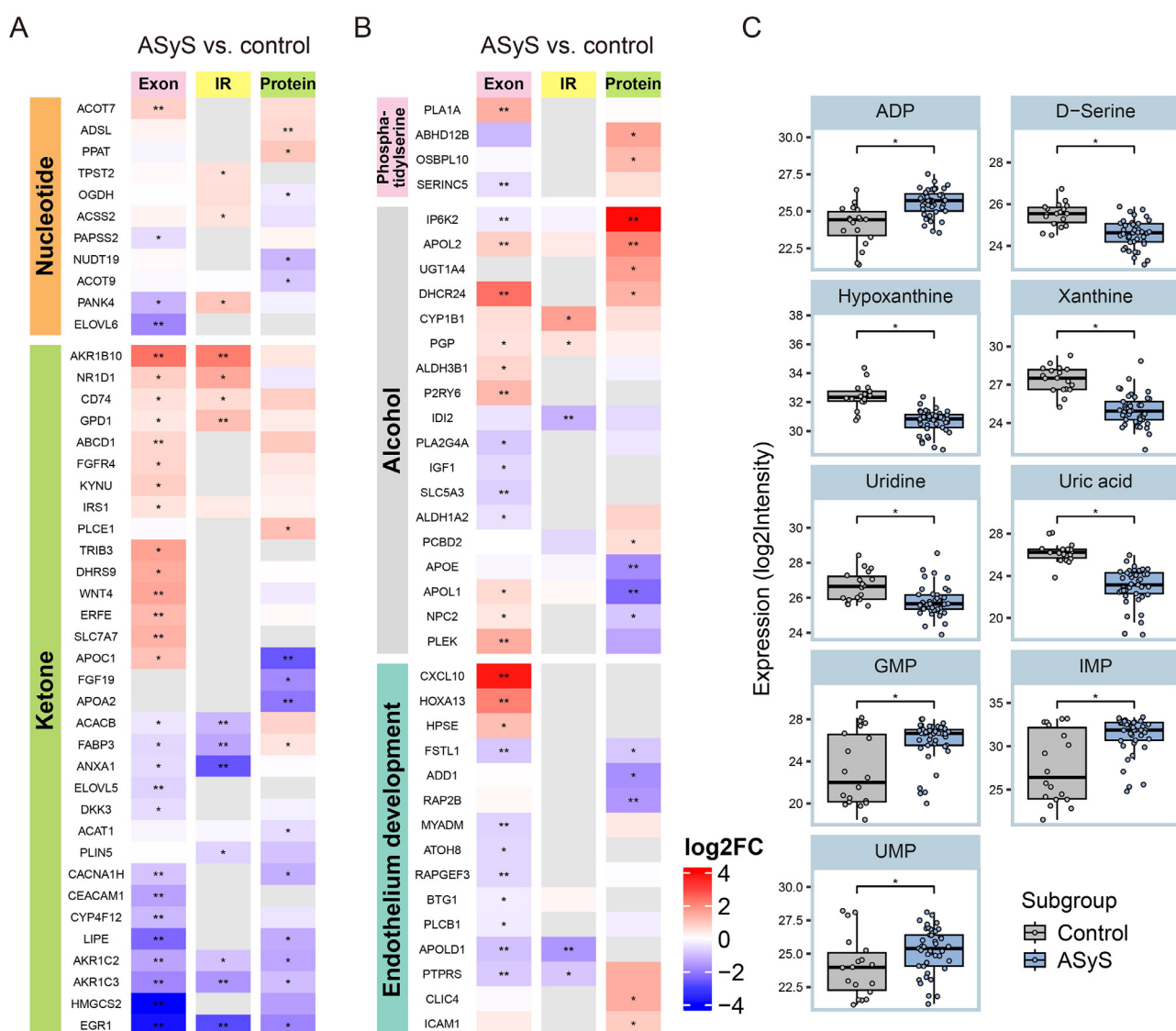


Figure 5. Metabolism and endothelial development pathways alterations in ASyS. (A) Heatmap showing the exon, IR, and protein expression of 11 genes in nucleotide metabolism, 32 genes in ketone metabolism. (B) Heatmap illustrating the exon, IR and protein expression of 4 genes in the phosphatidylserine pathway, 18 genes in alcohol metabolism, and 15 genes in endothelial development. The colour gradient represents log2(fold change), with red indicating higher expression. (C) Box plots showing altered expression of 9 metabolism-related metabolites in ASyS. * $P < .05$, ** $P < .001$ (ASyS vs control). ASyS, antisynthetase syndrome; IMP, inosine monophosphate; IRs, intron retentions; ADP, adenosine diphosphate; UMP, uridine monophosphate; GMP, guanosine monophosphate.

Additionally, significant alterations were observed at the protein level in critical metabolic enzymes such as phosphoribosyl pyrophosphate amidotransferase and adenylosuccinate lyase. These enzymes are involved in the purine biosynthesis pathway, converting phosphoribosyl pyrophosphate to inosine monophosphate (IMP), a precursor to uric acid. Moreover, metabolites such as D-serine, hypoxanthine, uric acid, uridine, and xanthine were significantly decreased in the ASyS group, while adenosine diphosphate (ADP), guanosine monophosphate (GMP), IMP, and uridine monophosphate (UMP) were increased (Fig 5C).

Endothelial dysfunction emerged as a key pathological process in ASyS. A total of 15 genes (11 in exons, 2 in IRs, and 5 in proteins) were enriched in the endothelium development pathway. The majority of these genes exhibited decreased expression across exons, IRs, and proteins, suggesting a deficiency in endothelial development in ASyS (Fig 5B). For example, *BTG1* and *FSTL1*, known regulators of endothelial proliferation, showed reduced expression at the exon or protein levels. These findings further emphasise the significant role of endothelial dysfunction in the pathology of ASyS.

All molecular changes in ASyS were categorised and illustrated in the corresponding pathway diagrams for clearer visualisation (Supplementary Figs S21, S22).

In the validation cohort, due to limited sample availability, metabolomics data were not available for ASyS. Nonetheless, metabolic pathway alterations and endothelial dysfunction were observed, showing 38.8% consistency at $P < .05$ vs control, with 68.8% of genes exhibiting consistent directional changes. The lower consistency in ASyS may be attributed to the small sample size ($n = 4$), despite the majority of genes showing similar expression trends (Supplementary Fig S23).

Pathway correlations and predictive biomarkers for IIM

We identified correlations between the activity scores of IIM subtype-specific pathways and clinical features. In DM, the IFN- α/β production pathways (across exon, IR, and protein levels) were positively correlated with skin manifestations, including heliotrope rash, Gottron's papules, V-sign, shawl sign, and holster sign, as well as with perifascicular atrophy. In contrast, the IFN- γ production score, primarily at the exon level, showed no correlation with skin rash but was positively associated with muscle enzyme levels (Alanine aminotransferase (ALT), aspartate aminotransferase (AST), lactate dehydrogenase (LDH), Creatine Kinase (CK), Myoglobin (Mb)), perifascicular necrosis, and inflammatory infiltration. Cytokine pathway showed a positive correlation with skin manifestations at the IR level, and with muscle enzyme, perifascicular atrophy, and inflammatory infiltration at the exon level (Fig 6).

In IMNM, cytoskeleton-related pathways (actin filament polymerisation, actomyosin structure organisation, and actin filament bundle assembly) and ECM-related pathways (ECM organisation, tissue remodelling, and connective tissue development) were negatively correlated with skin rash and MMT-8 scores. These pathways, however, were positively correlated with muscle enzyme levels, muscle fibre necrosis, regenerating fibres, inflammatory infiltration, and membrane attack complex (MAC) deposition, predominantly at the exon and IR levels (Fig 6).

In ASyS, alcohol metabolism (exon), ketone metabolism (exon), endothelium development (exon and protein), and metabolites (D-serine and uric acid) were negatively correlated with skin rash and positively correlated with muscle enzyme levels, muscle fibre necrosis, regenerating fibres, inflammatory

infiltration, and MAC deposition. Conversely, nucleotide and phosphatidylserine metabolism exhibited the opposite pattern at the IR level (Fig 6).

To evaluate the impact of IIM subtype-specific pathways on patient outcomes, we categorised patients into 2 groups based on disease relapse according to specific criteria [21,22]. Hazard ratios were calculated to evaluate the impact of these pathways on IIM prognosis. Higher scores for IFN- α/β production pathways (exon and protein levels), IFN- γ production (exon), cytokine-mediated signalling pathways (exon, IR, protein), actin filament bundle assembly (IR), tissue remodelling (IR), alcohol metabolism (exon and IR), ketone metabolism (exon), nucleotide metabolism (protein), phosphatidylserine metabolism (exon), endothelium development (exon and protein), and D-serine were associated with increased relapse risk. In contrast, higher levels of actomyosin structure organisation (IR), connective tissue development (protein), tissue remodelling (protein), ketone metabolism (IR), nucleotide metabolism (IR), and phosphatidylserine metabolism (protein) were linked to more favourable outcomes (Fig 7).

DISCUSSION

Integrative multiomics strategies, such as the MOFA used in this study, offer a powerful way to explore the intricate regulatory networks across transcriptomic, proteomic, metabolomic, and clinical data [13]. Here, we analysed muscle samples from 203 newly diagnosed patients with IIM and successfully identified distinct molecular signatures for each subtype. This approach addressed the limitations of single-omics data, as shown by PCA's inability to differentiate between subtypes, and demonstrated its ability to uncover hidden patterns within individual datasets. These findings underscore the value of integrative multiomics approaches in better understanding the complex nature of IIM, advancing insights into its pathogenesis, and identifying potential therapeutic targets and prognostic biomarkers.

The significant enrichment of IFN and cytokine signalling pathways in DM aligns with prior transcriptomic studies, particularly the well-established IFN gene signature [23,24]. Our findings of changes across exons, IRs, and proteins in genes such as *ISG15*, *MX1*, *STAT1*, and *DHX58* point to a complex regulatory network driving IFN overproduction. Type I IFN is associated with skin manifestations, such as heliotrope rash and Gottron's papules, as well as perifascicular atrophy, suggesting that type I IFN-driven inflammation primarily affects the skin and contributes to fibre atrophy. In contrast, IFN- γ and cytokine signalling pathways are more strongly linked to perifascicular necrosis and inflammatory cell infiltration. Elevated cytokines like *CCL19*, *CXCL9-11*, and *CCL13* in DM muscle tissues likely regulate leukocyte recruitment to the affected areas, indicating that IFN- γ and cytokine-driven inflammation primarily contribute to muscle inflammation [25]. Moreover, all DM-specific pathways were associated with an elevated relapse risk, implying that systemic immune activation influences IIM prognosis.

In addition, our scRNA-seq analysis of muscle tissues from patients with DM and ASyS identified distinct *ISG*^{hi} cell clusters, including T cells, myofibers, monocytes, and endothelial cells, characterised by high expression of *IFIT1*, *IFIT2*, *RSAD2*, and *ISG15* (data not shown in this study). These clusters were selectively expanded in DM, indicating widespread IFN-driven dysregulation across multiple cell types. Functionally, IFN activation contributes to muscle injury. Previous studies have highlighted the critical role of type I IFN in the initiation and progression of DM by amplifying inflammatory responses, activating

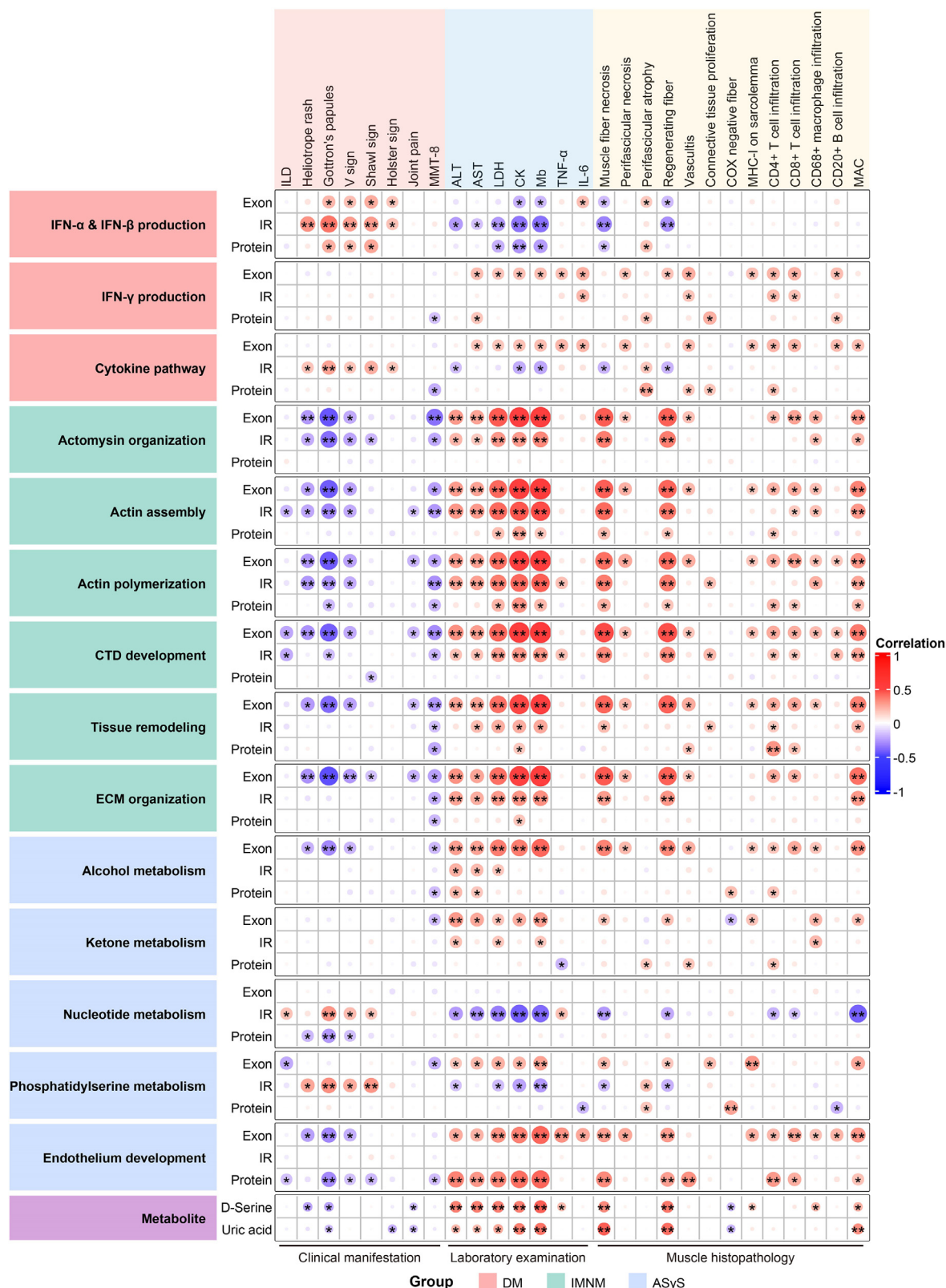


Figure 6. Correlations between IIM clinical characteristics, subtype-specific pathway activity scores, and metabolite expression. IIM clinical characteristics include interstitial lung disease (ILD), skin rash (heliotrope rash, Gottron's papules, V-sign, shawl sign, and holster sign), joint pain, Manual Muscle Test-8, muscle enzyme levels (ALT, AST, LDH, CK, Mb), inflammation activity (TNF- α and IL-6), and muscle histopathology (muscle fibre necrosis, perifascicular atrophy, perifascicular necrosis, regenerating fibres, vasculitis, connective tissue proliferation, cytochrome c oxidase [COX]-negative fibres, class I human major histocompatibility complex [MHC-I] on sarcolemma, CD20⁺ B cells, CD4⁺ T cells, CD68⁺ macrophages, CD8⁺ T cells, membrane attack complex [MAC]). Subtype-specific pathways include IFN- α and IFN- β production, IFN- γ production, and cytokine pathways in DM; actomyosin organisation, actin assembly, actin polymerisation, connective tissue development, tissue remodelling, and extracellular matrix (ECM) organisation in IMNM; alcohol, ketone, nucleotide, phosphatidylserine metabolism, and endothelial development in ASyS. Circle size and colour represent the magnitude of the correlation coefficient, with significance levels denoted by * $P < .05$, ** $P < .001$, calculated using Pearson, Spearman's rank, or point-biserial correlation. ASyS, antisynthetase syndrome; DM, dermatomyositis; IFN, interferon; IIM, idiopathic inflammatory myopathy; IMNM, immune-mediated necrotising myopathy; IL, interleukin; IRs, intron retentions; TNF, tumour necrosis factor; ALT, alanine aminotransferase; AST, aspartate aminotransferase; LDH, lactate dehydrogenase; CK, creatine kinase; Mb, myoglobin.

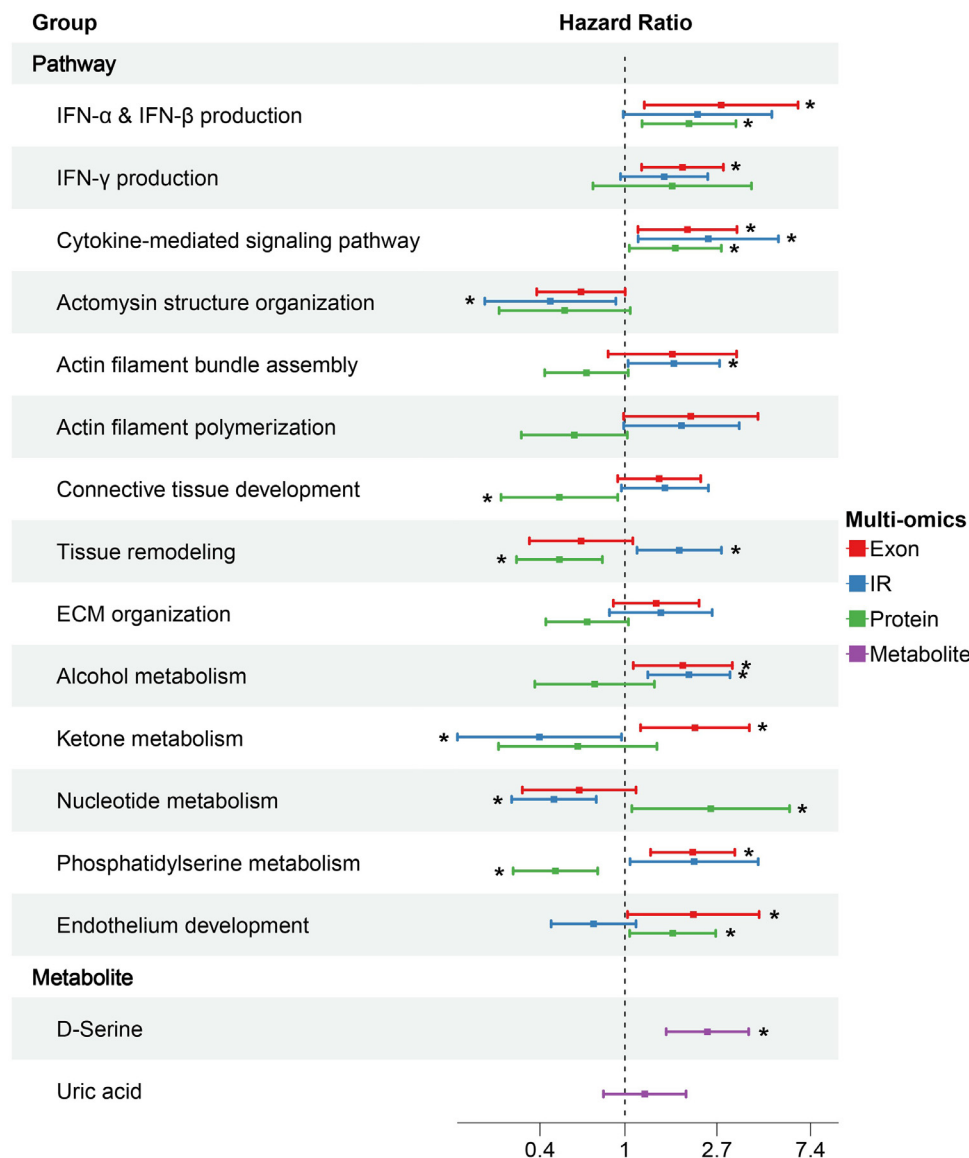


Figure 7. Forest plot showing the hazard ratio (HR) of IIM subtype-specific pathways on disease progression. Dots represent the HR, with error bars indicating the CI. Colours represent each omics. Statistical significance: * $P < .05$. ECM, extracellular matrix; IFN, interferon; IIM, idiopathic inflammatory myopathy; IRs, intron retentions.

endothelial cells, and interacting with MHC-I to induce mitochondrial dysfunction and elevate oxidative stress [26–29]. Similarly, IFN- γ promotes mitochondrial damage and oxidative stress in myositis, leading to cell death [30,31]. IFN- γ also impairs muscle regeneration by inhibiting the activation, proliferation, migration, and differentiation of myogenic progenitor cells, promoting their senescence via the JAK-STAT signalling pathway [32].

IMNM displays a unique molecular signature characterised by significant enrichment in cytoskeleton- and ECM-related pathways, reflecting structural damage and repair processes. Changes in the myosin gene family, actin-related genes, and troponin genes indicate disruptions in cytoskeletal integrity and muscle contractility. Additionally, all cytoskeleton- and ECM-related pathways are positively correlated with necrotic and regenerating muscle fibres, inflammatory infiltrates, and MAC deposition. However, while these pathways are markedly enriched in IMNM, they may represent consequences of muscle damage and regeneration rather than the primary drivers of IMNM pathology [33]. Many other significantly enriched pathways, such as myeloid, lymphocyte, and NK-T cell differentiation, macrophage migration, and interleukin-2 production,

highlight a strong immune-mediated component in the pathogenesis of IMNM. These pathways suggest that dysregulated immune cell activation and infiltration may play a central role in driving tissue inflammation and myofiber necrosis [34–36]. Notably, although tissue remodelling and connective tissue development (both at the protein level) were associated with a favourable prognosis in IMNM, other pathways showed no clear association with IIM outcomes, suggesting that muscle damage alone may not be the primary determinant of survival in IIM.

ASyS shows distinct metabolic alterations, particularly in ketone metabolism, extending its pathogenesis beyond previously reported plasma metabolite changes [37,38]. In healthy individuals, ketone metabolism enables rapid fuel switching during energy stress [39]. Its disruption is associated with metabolic, neurodegenerative, cardiovascular diseases, and cancer [40]. Ketone bodies serve as energy substrates and signalling/epigenetic regulators, contributing to mitochondrial dysfunction, oxidative stress, and inflammation [41]. For instance, β -hydroxybutyrate inhibits NLRP3 inflammasome and NF- κ B activation in macrophages, reducing proinflammatory cytokine production [42–44]. Although untargeted metabolomics did

not detect β -hydroxybutyrate in ASyS muscle tissue, expression of HMGCS2—the gene encoding the rate-limiting enzyme for ketone body synthesis (β -hydroxybutyrate)—was significantly reduced in ASyS [45]. This suggests disrupted ketone metabolism may indicate early molecular muscle injury ASyS.

Nucleotide metabolism is also disrupted in ASyS. Nucleotides are essential building blocks for life, serving primarily as precursors for nucleic acid synthesis to support cell proliferation [46]. They are produced through 2 main pathways: *de novo* synthesis and salvage. Beyond their structural functions, purine nucleotides such as cyclic adenosine monophosphate (cAMP), cyclic guanosine monophosphate (cGMP) also serve as crucial intracellular signalling molecules [47]. Imbalances in purine nucleotide levels can impair coordination between cell growth and division, making proliferating cells dependent on replication stress signalling [48]. Recent studies have uncovered broader, nonproliferative roles for purines and pyrimidines in regulating cellular function [49]. Both nucleotides and their breakdown products, nucleosides, have diverse effects. For example, ATP and ADP stimulate immune responses by activating natural killer cells, and increased purine levels can induce MHC-I polypeptide-related sequence A expression [50,51]. In contrast, nucleosides like adenosine have immunosuppressive effects, especially through regulatory T cells [52]. Nucleoside pool reprogramming also plays a key role in adaptation to oxidative and endoplasmic reticulum (ER) stress [53]. However, the role of nucleotide metabolism in muscle injury and the inflammatory microenvironment remains poorly understood. Notably, the enrichment of the alcohol metabolism pathway, despite the absence of heavy drinking history in our cohort, mirrored the proinflammatory and fibrotic features seen in alcoholic myopathy, pointing to shared underlying mechanisms [54].

This study has several notable limitations. First, the exclusion of IBM, due to its rarity in China, limits the generalisability of our findings to all IIM subtypes. Second, in our multiomics integration analysis (MOFA) of patients with IIM, we included only individuals with a single-positive MSA and excluded those who were autoantibody negative. This selection may have introduced bias in identifying molecular signatures specific to IIM subtypes. To mitigate this, we validated the reproducibility of these subtype-specific pathways in an independent cohort that included patients with MSA-negative IIM, which showed strong consistency with the original findings. Third, while MOFA is effective in integrating multiomics data, its reliance on muscle biopsies restricts insights into systemic manifestations, such as lung or skin involvement, which may involve distinct molecular pathways. Fourth, some control samples were obtained from the quadriceps femoris, while all IIM samples came from the biceps brachii. Despite anatomical differences, no significant changes in fibre type-specific genes were observed between the 2 muscles, minimising concerns about control sample heterogeneity in our cohort. Finally, functional studies, such as gene knock-downs or metabolomic interventions, are required to confirm the mechanistic roles of key factors like DHX58 and FABP3 in muscle injury.

In summary, our study offers a comprehensive and integrative multiomics characterisation of IIM, identifying subtype-specific molecular features and pathways and linking them to clinical manifestations and disease outcomes. By simultaneously analysing transcriptomic (exon and IR), proteomic, and metabolomic data from muscle tissue, we provide a multilayered perspective on disease heterogeneity. In DM, previous transcriptomic studies have demonstrated a strong association between IFN signalling and disease pathology [24]. Our analysis across

all 4 omics layers revealed robust activation of the IFN pathway. Specifically, we first identified changes in IFN-associated metabolites in DM muscle tissues. IFN- α/β signalling was more strongly associated with skin involvement, whereas IFN- γ production correlated with muscle damage and inflammatory infiltration. Both pathways were linked to an increased risk of relapse, underscoring their potential value as prognostic biomarkers. In IMNM, we found significant enrichment of cytoskeleton- and ECM-related pathways, which showed strong correlations with muscle fibre necrosis, regenerating fibres, and inflammatory infiltration. These changes likely reflect secondary tissue remodelling and repair rather than primary disease mechanisms. Whether enhancing these pathways could support muscle regeneration remains an important direction for future research. In ASyS, our study is the first to report abnormalities in multiple metabolic pathways and endothelial dysfunction within muscle tissue. While earlier research identified metabolic changes in peripheral blood mononuclear cells and plasma, our findings provide the first direct evidence of such alterations in the muscle microenvironment [37,38]. These pathways were positively associated with muscle fibre necrosis, inflammatory infiltration, and worse clinical outcomes, offering new insight into the role of local metabolism in tissue injury and repair. Together, this integrative multiomics analysis highlights distinct molecular pathways in IIM subtypes among Han Chinese, linking IFN and cytokine signalling to DM, cytoskeletal/ECM disruption to IMNM, and metabolic/vascular alterations to ASyS. These insights deepen our understanding of IIM pathogenesis and pave the way for tailored diagnostics and interventions, showcasing the transformative potential of multiomics in autoimmune research.

Competing interests

All authors declare they have no competing interests.

Acknowledgements

We would like to express our gratitude to all rheumatologists in the Department of Rheumatology and Immunology and Research Center for Neuroimmune and Neuromuscular disorders at Xiangya Hospital, Central South University, for their unwavering support and priceless clinical assistance throughout this study. Furthermore, we extend our gratitude to all patients and volunteers who participated in this study and provided us with biopsy samples.

Contributors

YX was responsible for conceptualisation, formal analysis, investigation, methodology, and funding acquisition. SX was responsible for conceptualisation, investigation, and writing - original draft. YL, YJ and HZ were responsible for resources. Hongdong L was responsible for methodology. XZ was responsible for resources and supervision. Hui L was responsible for Supervision, Funding acquisition. HZ was responsible for conceptualisation, supervision, writing - review and editing, and funding acquisition

Funding

This study was supported by grants from the Chinese National Key Technology R&D Program, Ministry of Science and Technology (2024YFC2510300); the National Natural

Science Foundation of China (82202002, 82471842, 82471840); the Natural Science Foundation of Hunan Province, China (2023JJ41002); and the Edith Busch Research Award from the World Scleroderma Foundation.

Patient consent for publication

Not applicable.

Ethics approval

The study involving human participants was reviewed and approved by the ethics committee of Xiangya Hospital, Central South University (201212074).

Provenance and peer review

Not commissioned; externally peer reviewed.

Data availability statement

Data are available on reasonable request.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ard.2025.08.011.

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